2c. Content of Our Natural World (J384/01)

The natural world contains a rich diversity of distinctive landscapes and ecosystems which are constantly changing through physical processes and human interactions. This component gives learners the opportunity to explore the natural world they live in, to understand why it looks the way it does and appreciate its value. It includes investigation of global hazards which humans face as well as an examination of how the climate is changing and what this means for the world today. Learners study the distinctive landscapes that surround them and the ecosystems that help sustain the life on Earth.

Topic 1 – Global Hazards

This topic allows learners to develop an understanding of a variety of hazards that impact human lives both within the UK and worldwide. Learners investigate how weather can be hazardous, gaining knowledge of the major processes within the atmosphere and their impact in creating extreme weather. This is contextualised through two case studies of natural weather hazard events. Earthquakes and volcanic eruptions are just some of the deadly hazards we face on Earth. Not only do they impact humans but they also shape our land. An understanding of tectonic hazards is developed; exploring the causes, consequences and responses to a tectonic event of choice.

1.1. How can weather be hazardous?		**Scale**
a. Why do we have weather extremes?	• Outline of the global circulation system including the effects of high and low pressure belts in creating climatic zones.	G, R
	• How the global circulation of the atmosphere causes extremes in weather conditions in different parts of the world. • The extremes in weather conditions associated with wind, temperature and precipitation in contrasting countries.	G, R
	• The distribution and frequency of tropical storms and drought, and whether these have changed over time. • Outline the causes of the extreme weather conditions associated with tropical storms. • Outline the causes of the extreme weather conditions of El Niño/La Niña leading to drought.	G, R
b. When does extreme weather become a hazard?	• **Case studies** of **two** contrasting natural weather hazard events arising from extreme weather conditions. The case studies must include a natural weather hazard from each bullet point below: ○ flash flooding or tropical storms ○ heat wave or drought. There must be **one** UK based and **one** non-UK based natural weather hazard event. • For each chosen hazard event, study the place specific causes (including the extreme weather conditions which led to the event), consequences of and responses to the hazard.	N, R, L, F

1.2. How do plate tectonics shape our world?		
a. What processes occur at plate boundaries?	• The structure of the Earth. • Understanding of slab pull and ridge push to show how tectonic plates move.	G
	• The processes that take place at constructive, destructive, conservative and collision plate boundaries as well as hotspots.	
	• How the movement of tectonic plates causes earthquakes, including shallow and deep focus, and volcanoes, including shield and composite.	
b. How can tectonic movement be hazardous?	• **A case study** of a tectonic event that has been hazardous for people, including specific causes, consequences of and responses to the event.	N, R, L
c. How does technology have the potential to save lives in hazard zones?	• How technological developments can have a positive impact on mitigation (such as building design, prediction, early warning systems) in areas prone to a tectonic hazard of your choice.	R, L